0.20% per annum, with high year-to-year variation and frequent setbacks was replaced by a much more steady growth rate of 1.5% per annum or better" (Mokyr, 2005, p. 1118).

The second phase in Figure 2 is followed by a third phase with the strongest growth rate of approximately 8% and which lasts to the start of the 21st century. Doubling time for cited references in this phase is approximately 9 years. The final phase has a negative growth rate. Around 1980, the curve in Figure 2 starts to flatten off and then fall significantly: from 1980, more publications are included in the regression analysis each year, which in many cases refer to publications which could not yet be cited in the previous year. The fourth segment therefore reflects "time-dependent database characteristics in, for instance, the registration of references" (van Raan, 2000, p. 348).

To establish in how far the use of all the publications since 1980 has an influence on the growth rate of science (particularly in the last 30 years), we have extracted the annual growth rates of the cited references given in the publications for 2012 in Figure 3. 1,859,648 publications (from 2012) and 53,345,550 cited references between 1650 and 2012 were included in this analysis. As illustrated by a comparison of the results in Table 3 with the results in Table 2 the regression analysis identified four segments with different growth rates even though only one year was taken into account. While the growth rates for the medium and younger segment hardly differ, (we are looking here at just the segments with a positive growth rate), when only one year is taken into account, the growth rate is much lower in the oldest segment than in the analysis of citing publications from 1980 to 2012. By taking just one year into account, the number of cited references particularly in this segment falls as generally speaking, very old publications are rarely cited (Marx, 2011). In contrast to Figure 2 we see in Figure 3 an abrupt decline in the number of cited references from around 2010. van Raan (2000), who observes a similar fall a few years before 1998 in his data, explains this drastic decline thus: "It is well known, particularly in the natural science and medical fields, that publications of a given year (here: 1998) have a peak in the distribution of their references around the age of three years. Therefore, [in the figure] in the last three years the number of references will decrease” (p. 350).

According to van Raan (2000) the growth rates in Figure 2 and Figure 3 reflect two processes: "*ageing* (scientists will be increasingly less interested in increasingly older literature), and *growth* (there were simply much less papers published in 1930 as there are in 1990, so there is less to be cited to earlier years) " (p. 351). Tabah (1999) also discusses these two processes in connection with growth in science; they influence each other as described here: "Specifically, the faster a literature or even a given journal grows (in terms of number of articles published per year), the more rapidly it ages" (p. 250, see also Egghe & Rousseau, 2000). Similarly van Raan (2000) writes: "Although undoubtedly ageing of earlier published work is part of reality, another part of reality is that in earlier times there were many fewer documents published than in recent times. Thus, the time-dependent distribution of references will always be a specific combination of the ageing and growth phenomena of science, or better said: of scientific literature" (pp. 347-348).

In earlier analyses of the growth rate of science, the time before, between and WWI and WWII has been of particular interest (Larsen & von Ins, 2010; Price, 1965; van Raan, 2000). In conformity with our results, other studies have also shown a significant fall in the number of cited references during this period. Furthermore, this and other studies, as van Raan (2000) writes, also show that the two dips are "immediately followed by an astonishing ‘recovery’ of science after the wars toward a level more or less extrapolated from the period before the wars" (p. 347). However the results for the period between WWI and WWII, from around 1920 to 1935, revealed some inconsistencies: "In Price’s analysis this period shows a ‘hill’ when extrapolating the exponential increase of the pre-World War I period. This ‘hill’ has almost disappeared in our analysis" (van Raan, 2000, p. 349). Like Price's, our study shows that a significant increase in the number of cited references in the period between the wars has several consequences, one of which is that the third segment with a growth rate of around 8% (see Table 2) does not begin in the 'Big Science' period (see above), but a few years earlier. According to our data, WWII is only a temporary low point in the significant growth of science since the period between the wars.

In our final analysis we looked at the extent to which we could determine whether there were differences between growth rates in different disciplines. The main categories of the Organisation for Economic Co-operation and Development (2007) (OECD) were used as a subject area scheme for this study. A concordance table between the OECD categories and the WoS subject categories is provided by InCites (Thomson Reuters) (see http://incites-help.isiknowledge.com/live/globalComparisonsGroup/globalComparisons/subjAreaSchemesGroup/oecd.html). The OECD scheme enables the use of six broad subject categories for WoS data: (1) natural sciences, (2) engineering and technology, (3) medical and health sciences, (4) agricultural sciences, (5) social sciences, and (6) humanities. However, neither the numbers for engineering and technology nor those for the social sciences and humanities were included in this study. According to the Council of Canadian Academies (2012), the usefulness of citation impact indicators depends on the extent to which the research outputs are covered in bibliometric databases, and this coverage varies by subject category. The coverage tends to be high in the natural and life sciences, which place a high priority on journal publications. In engineering, the social sciences and humanities, where the publication of books, book chapters, monographs, conference proceedings etc. is more traditional, the extent of the coverage is reduced (Lariviere, Gingras, & Archambault, 2006).

We therefore undertook discipline-specific analyses for the natural sciences and the medical and health sciences. 15,435,641 publications from 1980 to 2012 and corresponding 379,294,777 cited references from 1650 to 2012 were included in the analyses for the natural sciences. The corresponding figures for the medical and health sciences were 12,796,558 publications and 256,164,353 cited references. The results of the analyses are shown in Figure 4, Figure 5, Table 4, and Table 5. The results are very similar. The regression analysis identified 3 segments with interpretable content up to the beginning of the 21st century, with similar start and end points. Only the middle segment, which exhibits a higher level of scientific activity compared to the oldest segment, begins slightly earlier in the natural sciences than in the medical and health sciences (in 1720 rather than 1750). Furthermore, the growth rates in the medical and health sciences since the 18th century have been minimally higher than in the natural sciences. The doubling times in the medical and health sciences are correspondingly slightly lower than in the natural sciences.

The results of the regression analysis in which the interaction terms with disciplines are included confirm the similarities. The interaction effects (results are not shown here) disciplines × publication years for each segment are statistically significant, but they are too small to justify any statements about real differences between the disciplines. The statistical significance results (more or less) from the high proportion of explained variance and the associated very low standard errors of the parameters.

# 4 Discussion

In this study we have looked at the rate at which science has grown in terms of number of publications and cited references since the mid-1600s. In our analysis we identified three growth phases in the development of science, which each led to growth rates tripling in comparison with the previous phase: from less than 1% up to the middle of the 18th century, to 2 to 3% up to the period between the two world wars and 8 to 9% to 2012. As the growth rate of the cited references in the third segment is significantly higher than that based on the source segments in this study (approximately 3%), we can assume that WoS only covered a small part of the total publications. We can further suppose that the number of early publications is underestimated and the publications' growth rate is overestimated in the cited reference analysis because of *aging* (see above). van Raan (2000) reports, based on analyses of cited references, a growth rate of around 10% for the period from 1800 to the mid-1990s.